Fizika

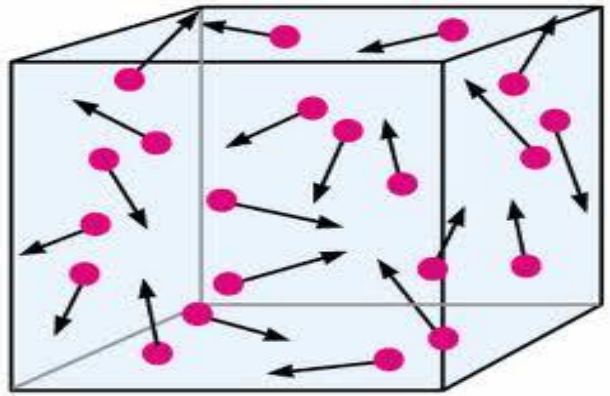
9 sinf

MAVZU: Modda tuzilishining molekulyar – kinetik nazariyasi.



Molekulyar-kinetik nazariya (MKN)

Moddaning tuzilishi va xossalarini uni tashkil qilgan molekulalarning harakatiga hamda molekulalar orasidagi o‘zaro ta’sir kuchining mavjudligiga bog‘lab o‘rganuvchi nazariya molekulyar-kinetik nazariya (MKN) deb ataladi.



Molekulyar-kinetik nazariya qoidalari

MKN tajribalarda isbotlangan uchta qoidaga asoslangan.

1. Moddalar zarralardan – atom va molekulalardan tashkil topgan.

2. Atom va molekulalar to‘xtovsiz va tartibsiz harakat qiladi.

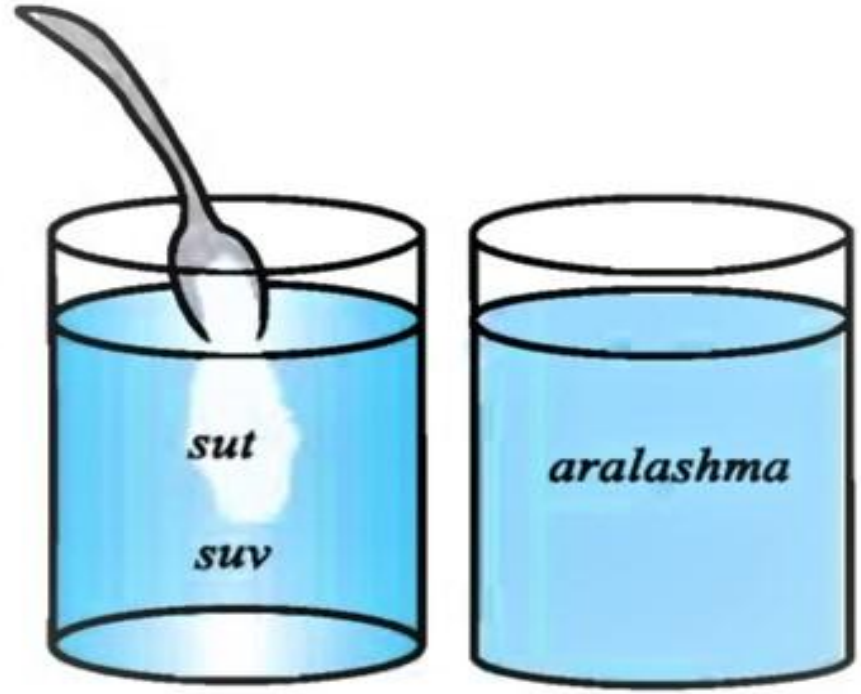
3. Atom va molekulalar orasida o‘zaro tortishish va itarilish kuchlari mavjud.

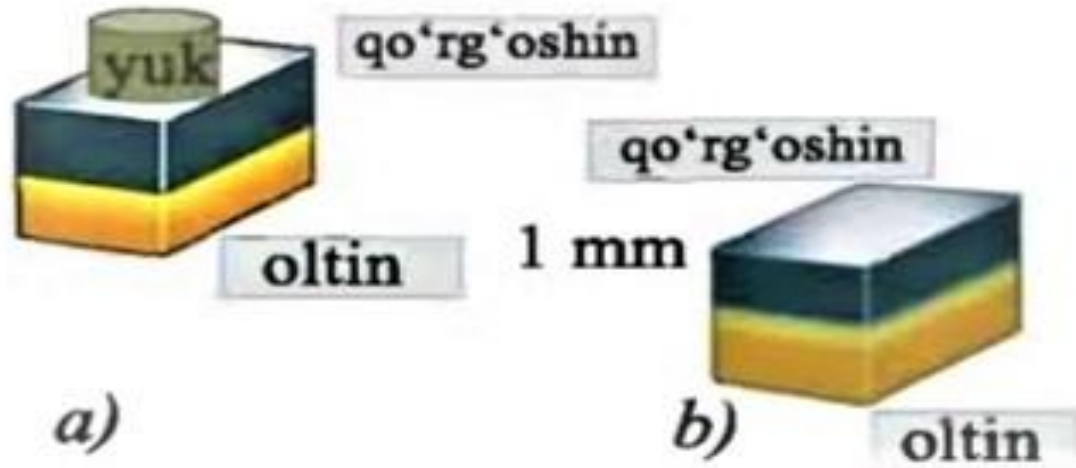
Tajribalar

1. Gazlarda: Xonaning bir chekkasida atir sepilsa, bir oz vaqtdan so‘ng atir xidi xonaning boshqa chekkalariga ham tarqaladi.

2. Suyuqliklarda: Stakandagi suv ustiga qoshiqda sut solinadi. Suv va sut asta-sekin aralashadi. Aralashganida vaqt ketishi esa zarralarning o‘zaro harakatlanishini ko‘rsatadi.

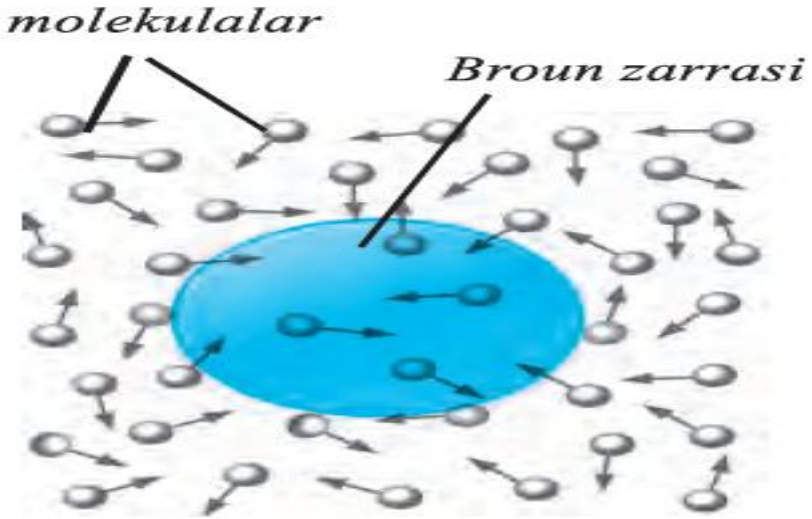
3. Qattiq jismlarda: Qattiq jism zarralari juda sekin aralashadi. Bunga sabab qattiq jism zarralari o‘zaro kuchli tortishadi.


sut
suv
aralashma
a)
b)


yuk
qo‘rg‘oshin
oltin
a)
qo‘rg‘oshin
1 mm
b)
oltin

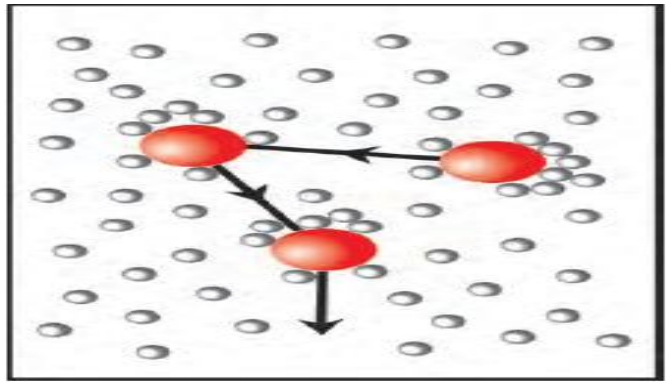
Broun harakati

Suyuqlik yoki gazdagi atom va molekulalarning to‘xtovsiz va tartibsiz harakati xaotik harakat yoki Broun harakati deyiladi.

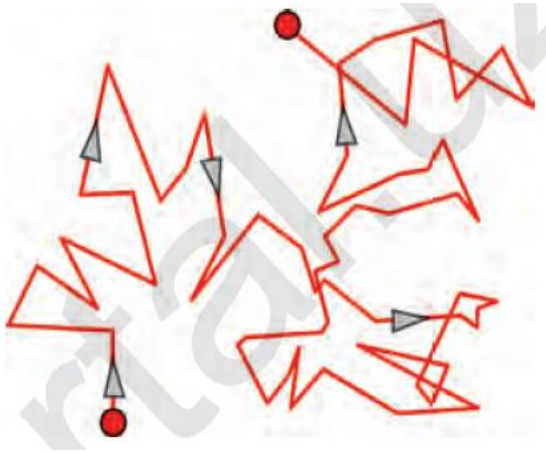


Broun harakati

1905- yilda Albert Eynshteyn tomonidan Broun harakatining molekulyar - kinetik nazariyasi yaratildi.



Broun harakatini fransuz fizigi Jan Perren tajribada o‘rganib, xaotik harakat qilayotgan zarraning teng vaqtlar oralig‘idagi holatlarini suratga olgan.



Amaliy topshiriq

Mexanik model asosida molekulalarning tartibsiz harakatini ko‘rsatish.

Kerakli jihozlar: ikki xil rangli sharlar, tarelka, flomaster.



a)



b)

UYGA VAZIFA

1. Molekulyar-kinetik nazariya qanday qoidalardan iborat?
2. Broun harakatining yuzaga kelish sababini yozing.
3. Amaliy topshiriqni bajaring va xulosa yozing.(7- bet).